Midterm Report on Decadal Mission — 2026: Towards the New Physics Higgs/Flavour Era

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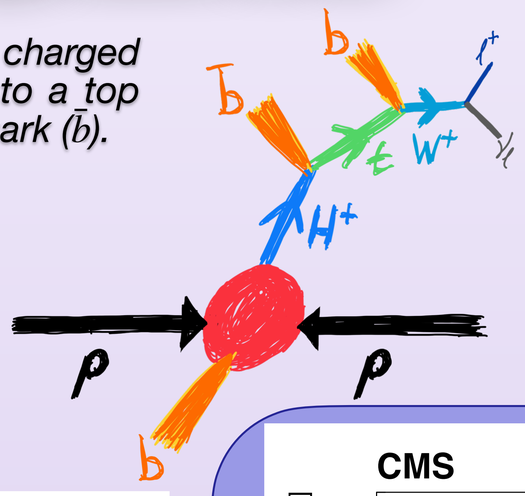
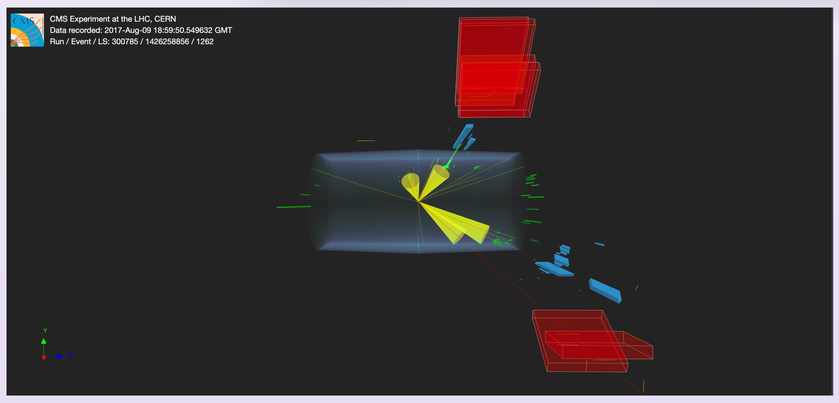
- No hints of new physics beyond Standard Model (SM) to understand unexplained phenomena: **matter-antimatter imbalance in the Universe, missing-mass (dark matter) problem, heaviness of the top quark.**
- No Limitations in the SM that prevent multiple Higgs bosons, including electric charged ones.
- ***We believe energy scales best suited for LHC (i.e. < 1 TeV), and rare processes that we could still probe haven't been exhausted*** —> Look for extra Higgs bosons and their couplings at a mass scale of ~500 GeV with independent measurements based on a **generalised ver sion of the two-Higgs-doublet model (g2HDM)** [1].
- 2HDM is a model with four new Higgs bosons: A, H, H^+, H^- with new couplings of the Higgs bosons to SM particles.
- **Our first experimental measurement** in this direction searched for A and H bosons: observed no significant excess for new sub-TeV Higgs bosons above SM predictions with ~ 1 extra Yukawa couplings [2,3].
- **Next step in our experimental search program is to search for charged Higgs bosons [4] inspired by g2HDM.**

Search for charged Higgs bosons via the $pp \to (q)H^{\pm} \to (q)tb$ process in proton-proton collisions at $\sqrt{s} = 13$ TeV

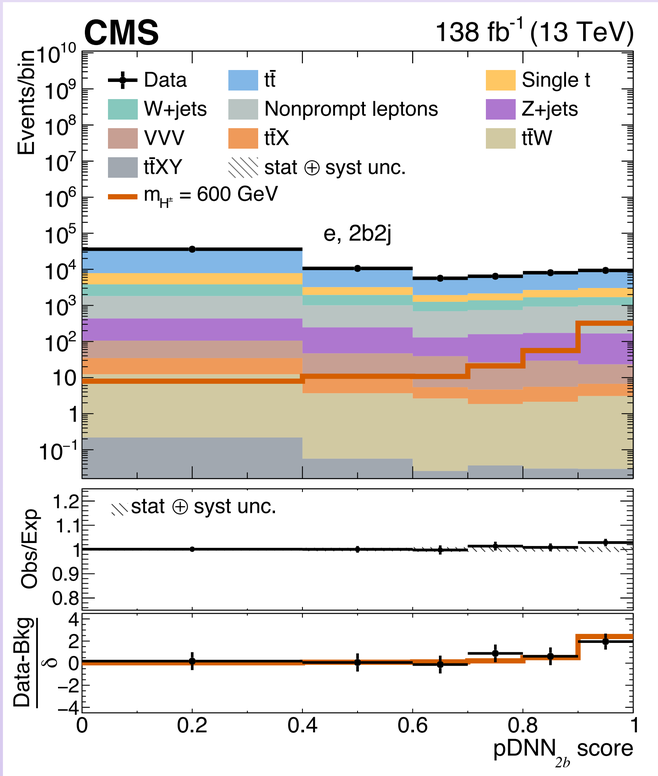
Production of an electrically charged Higgs boson (H^+) decaying to a top quark (t) and a bottom antiquark ($\bar{b}$).

Experimental signature: a high p_T lepton (ℓ), b-quark jets (b), light jets (j), missing energy (ν).

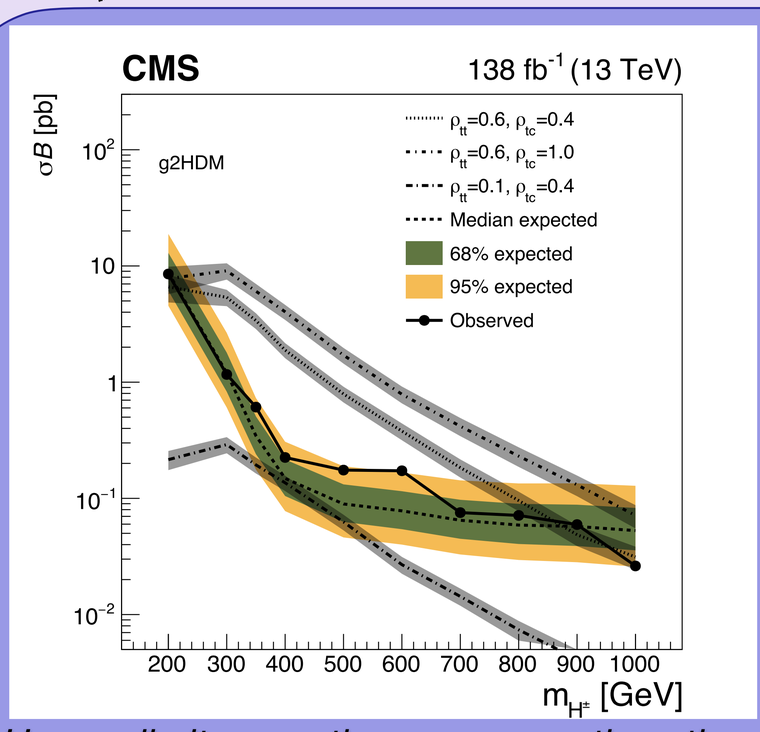




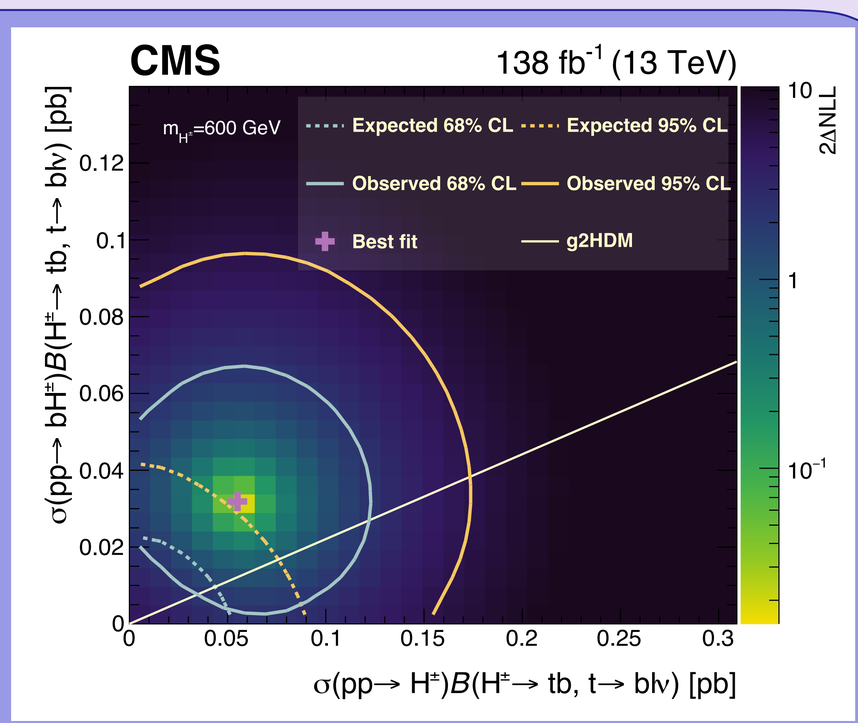
A candidate signal event as reconstructed in the CMS detector.



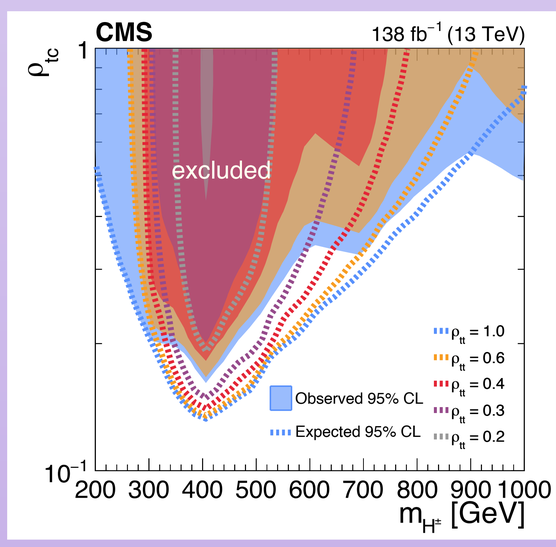
- *Signal regions (SRs) defined based in number of b-tagged jets and jets.*
- *Parametric deep neural network (pDNN) distributions to separate signal and background events.*



Upper limits on the cross section times branching ratio vs mass of the $H^{\pm}$. Theoretical predictions with different extra Yukawa couplings (ρ_{tc} & ρ_{tt}) shown with grey bands.



- *Best fit cross section times branching fraction for $m_{H^{\pm}} = 600$ GeV:*
  $\sigma(pp \to H^{\pm})\mathscr{B}(H^{\pm} \to tb, t \to b\ell\nu) = 0.055^{+0.068}_{-0.055}$ *pb*
- ***An excess is observed w.r.t. SM expectation with a local significance of 2.5 standard deviations (σ) for a signal with $m_{H^{\pm}} = 600$ GeV.***
- ***The excess looks like signal both shape- and cross section-wise.***



Excluded phase-space regions vs $H^{\pm}$ mass & extra Yukawa coupling ρ_{tc} for various ρ_{tt} values presented with different colours.

- This is the first ever search of a charged Higgs boson based on the g2HDM.
- We continue to search for additional Yukawa couplings in Run 3.
- If the deviation at $m_{H^{\pm}} = 600$ GeV is real, the expected significance with full Run 3 data is ~4σ and with the combined Run 2 + Run 3 data, it is $\gtrsim 4.5\sigma$
- HL-LHC data with much higher luminosity will be valuable to push the frontier in the searches for additional Higgs bosons.

References
[1] D.K. Ghosh, W.-S. Hou, T. Modak, *Phys. Rev. Lett. 125 (2020) 221810*
[2] CMS Collaboration, *Phys. Lett. B 850 (2024) 138478*
[3] CMS Collaboration, *https://cms.cern/news/hunting-higgs-boson-siblings-top-quarks*
[4] CMS Collaboration, *CMS-B2G-24-008, arXiv:2512.2447, submitted to JHEP*

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